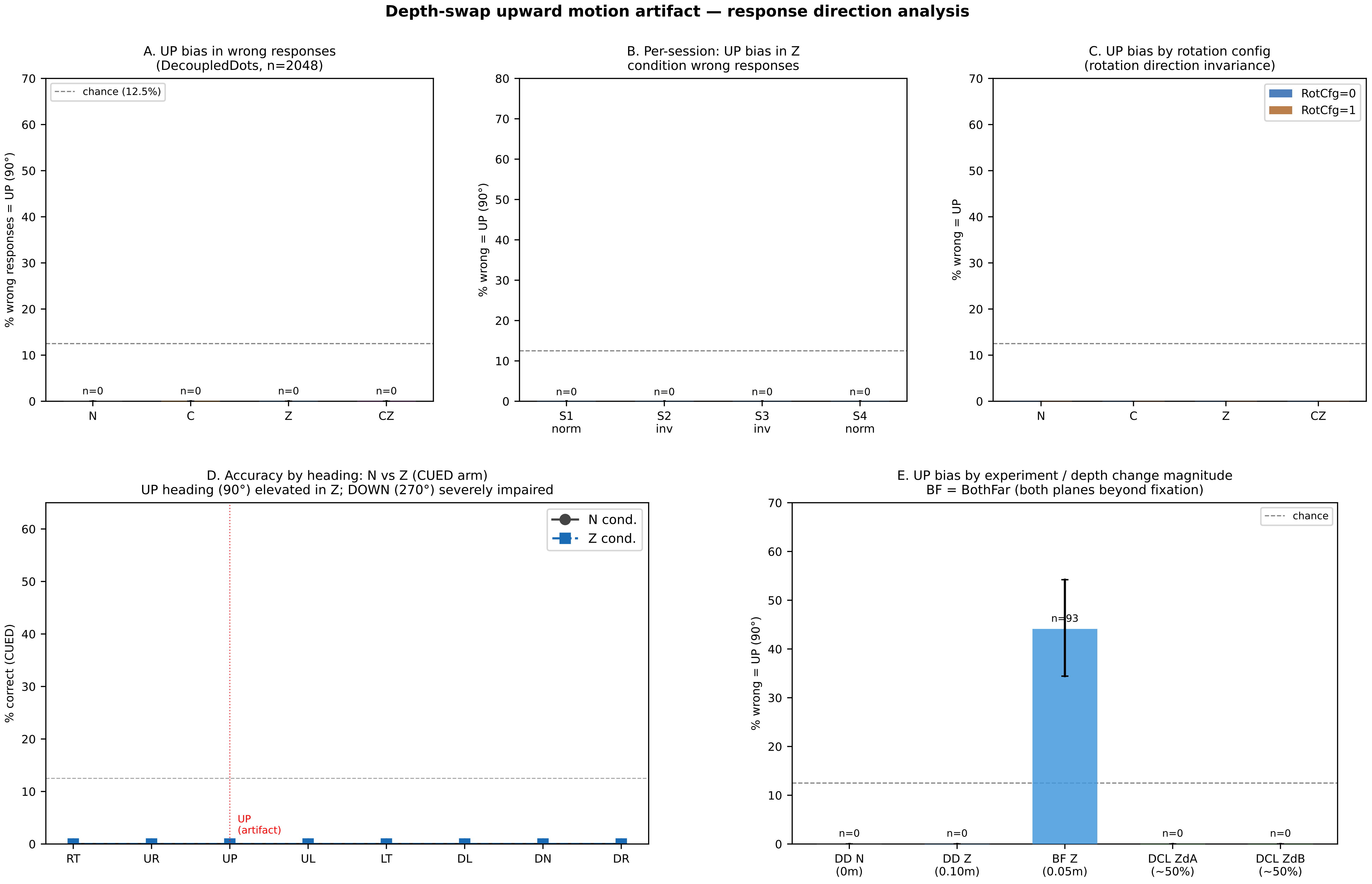
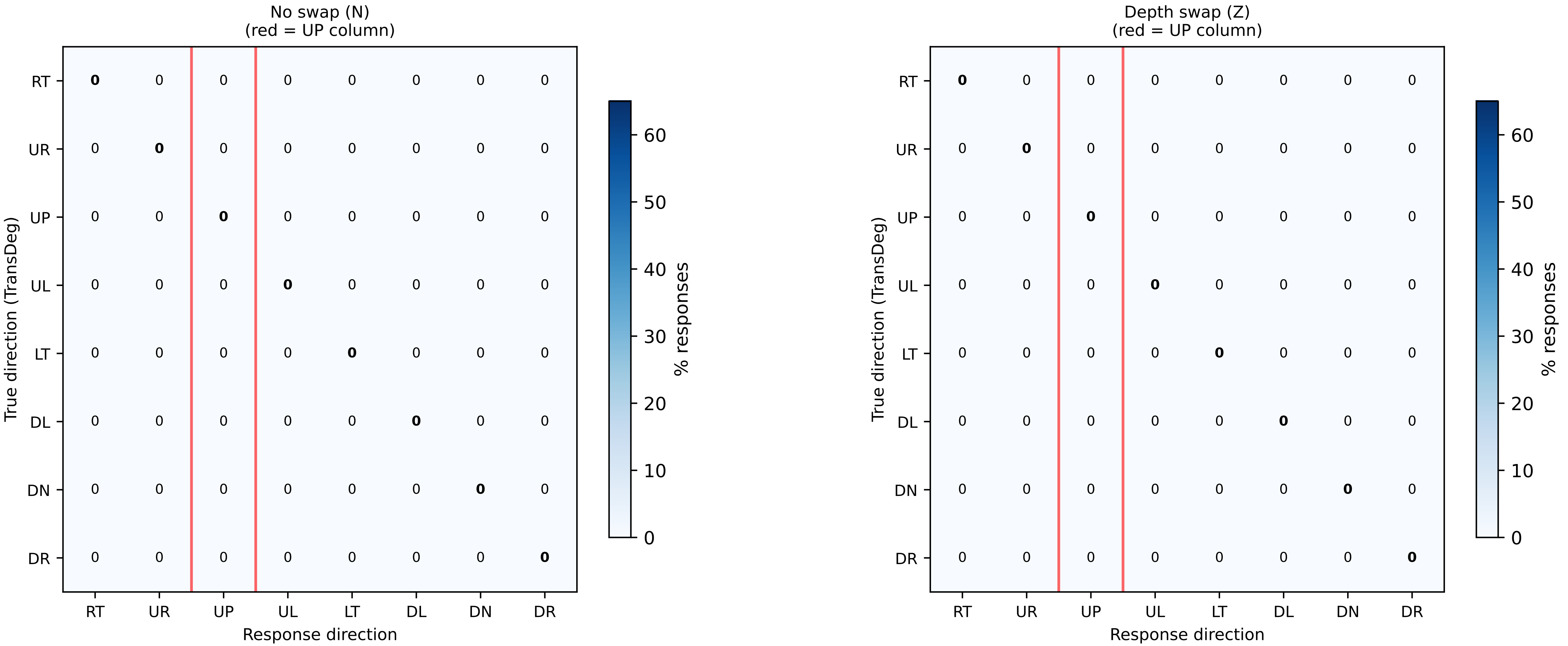


Depth-swap upward motion artifact — response direction analysis
A. UP bias in wrong responses
(DecoupledDots, n=2048)
chance (12.5%)
% wrong responses = UP (90°)
n=0
N
C
Z
CZ
B. Per-session: UP bias in Z
condition wrong responses
% wrong = UP (90°)
S1
norm
S2
inv
S3
inv
S4
norm
C. UP bias by rotation config
(rotation direction invariance)
RotCfg=0
RotCfg=1
% wrong = UP
D. Accuracy by heading: N vs Z (CUED arm)
UP heading (90°) elevated in Z; DOWN (270°) severely impaired
N cond.
Z cond.
% correct (CUED)
UP
(artifact)
RT
UR
UP
UL
LT
DL
DN
DR
E. UP bias by experiment / depth change magnitude
BF = BothFar (both planes beyond fixation)
chance
% wrong = UP (90°)
n=93
DD N
(0m)
DD Z
(0.10m)
BF Z
(0.05m)
DCL ZdA
(~50%)
DCL ZdB
(~50%)

Response confusion matrices — N condition vs Z condition (DecoupledDots)



Depth-swap artifact: mechanism and impact on results

MECHANISM

When depth swap occurs at tStart (Z and CZ conditions), all 4 subfields simultaneously change depth-plane assignment. StimulusBuilder.ApplyDepthOffsets() then shifts each dot by ±depthOffset_m along transform.forward. If transform.forward is not aligned with the camera's optical axis (e.g., because the StimulusBuilder is pitched slightly upward in world space while the camera is looking straight ahead or slightly downward), this depth change has a component in the screen-vertical direction.

For a pitch angle $\theta \approx 5–10°$, a 0.10m depth change produces:
$\Delta y\_screen = 0.10 \times \sin(\theta) / 2.0m \approx 0.25–0.5°$ apparent upward displacement
Delivered in 1 frame (~13ms at 75Hz) → apparent velocity ≈ 20–40 deg/sec
Translation signal = 2.26 deg/sec over 80ms = 0.18° total
→ The artifact impulse is ~100–200× faster than the signal and precedes it.

EVIDENCE

- UP (90°) share of wrong responses: N=4%, C=7%, Z=50%, CZ=51% (DD, n=2048)
- Consistent across sessions (37–59% per session), RotCfg 0 vs 1 (52% vs 47%), CUED vs UNCUED (51% vs 49%), DelayedFieldDepth N vs F (48% vs 52%)
- For TransDeg=270 (DOWN): Z condition produces 60.9% UP responses → observer correctly perceives motion, but artifact overrides direction
- For TransDeg=90 (UP): Z condition accuracy = 45.3% vs N = 35.9% (artifact HELPS)
- BothFar (depth change = 0.05m): UP bias = 44% (reduced vs 50% but still large)
- DCL ZdA/ZdB (50% of dots swap): UP bias = 26% (further reduced)
- Bias absent in N, C conditions (no depth swap) → definitively tied to ApplyDepthOffsets

IMPACT ON CUEING RESULTS

- Cueing advantage (CUED − UNCUED Δ) is PRESERVED: artifact affects both arms equally.
  Z: CUED UP-wrong = 51%, UNCUED UP-wrong = 49% — nearly identical.
  → The depth continuity effect on cueing is real, not an artifact.

- Absolute accuracy in Z/CZ is underestimated for non-UP headings and overestimated for the UP heading. The average across all 8 headings is deflated because UP trials (1/8 of all trials) have inflated accuracy that does not offset the 7/8 deflated.

- The apparent magnitude of depth disruption may be inflated: the Z→N accuracy drop includes both (a) the true depth-identity disruption and (b) the artifact's misdirection. These are impossible to separate at current data without heading-stratified analysis.

PROPOSED FIX

In StimulusBuilder.ApplyDepthOffsets(), replace:

```
Vector3 zVec = transform.forward * z;
```

with:

```
Vector3 camFwd = Camera.main != null ? Camera.main.transform.forward : transform.forward;
Vector3 zVec = camFwd * z;
```

This ensures depth offsets are applied along the camera's actual optical axis regardless of the StimulusBuilder's world-space orientation. Should be tested in-headset before collecting new data — verify that the depth percept (Near/Far separation) is preserved.